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4 May 1959
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MILITARY SPECIFICATION

CRADLE, BOMB HANDLING MHU-19/E

1. SCOPE.- This specification establishes the requirements for one type of bomb handling cradle designated as MHU-19/E, hereafter referred to as the cradle. (See 6.1)

2. APPLICABLE DOCUMENTS.- The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent required by this specification:

SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Waterproofed Kraft
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification For
MIL-T-5021	Tests, Aircraft Welding Operator's Certification
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-D-6843	Electrodes, Welding, Covered, Mild and Alloy Steel (For Aircraft Applications)
MIL-L-7711	Lubricating Grease (General Purpose Aircraft)
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification For
MIL-E-8697	Electrodes, Welding, Covered, Heat Treatable Steel
MIL-T-9107	Test Reports, Preparation Of
MIL-W-13748	Welding, Metal-Arc; Steels, Constructional Readily Weldable for Highly Stressed Joints

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STANDARDS

Military

MIL-STD-22	Welded Joint Designs
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Standard Manufacturing Company

T1-W1-J	Assembly, Cradle, MHU-19/E

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 DESIGN AND CONSTRUCTION.- Design and construction of the cradle shall be in conformance to Standard Manufacturing Company drawing T1-W1-J and subsidiary drawings.

3.2.1 INTERCHANGEABILITY.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.2.2 WELDING.- All welding shall be in accordance with MIL-W-13748 and MIL-STD-22. Unless otherwise indicated on detail drawings, the welding rod shall be in accordance with MIL-E-6843, Class A for sheet metal parts and MIL-E-8697, Class E-10016 for all parts other than sheet metal.

3.2.2.1 INSPECTION.- During fabrication of the cradle, all welding shall be inspected by personnel certified in accordance with MIL-T-5021.

3.3 PERFORMANCE REQUIREMENTS

3.3.1 PROOF LOAD.- The cradle, when in normal transport position, shall be capable of sustaining a 30,000 pound test load applied to the supports without failure or permanent deformation.

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3.3.2 ROADABILITY.- The cradle, carrying a 10,000 pound load, and loaded on the MHU-7/M Trailer shall satisfy the type II mobility requirements of MIL-M-8090.

3.3.3 MOBILITY.- The cradle, operating as an independent vehicle, while carrying a 10,000 pound load, shall satisfy the type I class 1 mobility requirements of MIL-M-8090 and in addition thereto be capable of backing at speeds up to 5 mph and abrupt backward starting.

3.3.4 CASTERING.- The cradle shall be equipped with four casters. The forward casters shall be capable of 360 degree rotation and the rear casters shall be capable of a minimum of 180 degree rotation. All casters shall be capable of locking in the forward transport position and in a position 90 degrees to the forward transport position.

3.3.5 COMPATIBILITY.- The cradle shall be capable of supporting the MHU-20/C Clip-In Subassembly unloaded, partially loaded or fully loaded while the assembly is being loaded or unloaded into the aircraft. The cradle shall have sufficient clearance between the frame and the lower store of the MHU-20/C Clip-In Subassembly so that the weapons can be loaded into the MHU-20/C when it is unloaded and supported on the cradle. The cradle shall have pick-up points which are compatible with the Trailer, Bomb MHU-7/M, the Boeing "Running W" Hoist Tube; Hoist Tube Set, 24,000 Lb Capacity, Type MB-1; Truck, Straddle, Special Weapon, 50,000 Lb Capacity, Type MH-1 and Truck, Straddle, Special Weapon, 30,000 Lb Capacity, Type MH-2.

3.3.6 AIR TRANSPORTABILITY.- The cradle, without load, shall be capable of withstanding the static loads encountered during air transportation.

3.3.7 ENVIRONMENTAL REQUIREMENTS.- The cradle shall be capable of satisfactory performance after being subjected to the following environmental conditions:

a. Low temperatures down to -54°C (-65°F)

b. Icing conditions such as may be encountered in arctic operation.

c. High temperatures up to 71°C (160°F)

d. Sand and dust conditions comparable to those of desert operation.

e. Salt atmosphere comparable to sea coast operation and shipment over salt water.

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3.4 IDENTIFICATION OF PRODUCT.- The cradle shall be marked for identification in accordance with MIL-STD-130.

3.5 GOVERNMENT LOANED PROPERTY.- The following items of equipment shall be loaned to the contractor, upon his request, as government loaned property for the purpose of testing the cradle:

Item No.	Type or Designator	Stock No.	Item Description	Quantity
1	MHU-7/M		Trailer, Bomb	1
2	MHU-20/C		Clip-In Subassembly	1
3	MB-1	1190-333-9614	Hoist Tube Set, 24,000 Lb Capacity	1

3.6 WORKMANSHIP.- The cradle shall be fabricated and finished in a manner that will insure that the criteria of appearance, fit, and adherence to specified tolerances are observed. These criteria must apply to all accessories and to all external non-functioning parts; e.g., cases, covers, and nameplates, as well as to functional parts. Particular attention shall be given to the neatness and thoroughness of marking of parts and assemblies, plating, painting, machine screw assemblage, welding and freedom of parts from burrs and sharp edges which may mar associated equipment or cause injury to operating personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the cradle shall be as shown in table I.

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES.- The preproduction samples shall consist of models representative of the production cradle. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.2 PREPRODUCTION TEST REPORT.- In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report conforming to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

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TABLE I

TESTS	REQUIREMENT	TEST PARAGRAPH	PREPRODUCTION	PREPRODUCTION ONLY	ACCEPTANCE	
					INDIVIDUAL	SAMPLING
Examination of Product	3.2; 3.6	4.5.1	X		X	
Conformance to Drawings	3.2; 3.6	4.5.2	X		X	
Proof Load Test	3.3.1	4.5.3	X			X
Road Test	3.3.2	4.5.4	X			X
Mobility Test	3.3.3	4.5.5	X			X
Castering Test	3.3.4	4.5.6	X			X
Fit Test	3.3.5	4.5.7	X	X		
Flight Load Test	3.3.6	4.5.8	X	X		
Low Temperature Test	3.3.7	4.5.9	X	X		
Icing Test	3.3.7	4.5.10	X	X		
High Temperature Test	3.3.7	4.5.11	X	X		
Sand and Dust Test	3.3.7	4.5.12	X	X		
Salt Spray Test	3.3.7	4.5.13	X	X		

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4.2.3 PREPRODUCTION TESTS.- Preproduction tests shall consist of all tests specified in this specification.

4.3 ACCEPTANCE TESTS.- Acceptance tests shall consist of the following:

a. Individual Tests
b. Sampling Tests

4.3.1 INDIVIDUAL TESTS.- Individual tests shall be as shown in table I.

4.3.2 SAMPLING TESTS.- Sampling tests shall be as shown in table I.

4.3.2.1 SAMPLING PLAN.- The sampling plan and sample size shall be as follows:

Production Lot	Number of Samples
First 50	1
Next 50	1
Each succeeding 100 or portion thereof	1

4.3.2.2 REJECTION AND RETEST.- When the sample item fails to satisfy the specified tests, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.3.2.3 INDIVIDUAL TESTS MAY CONTINUE.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or produced later shall not be made until it is determined that the items meet all requirements of this specification.

4.3.4 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 TEST CONDITIONS.- Test conditions shall be as described in each specific test in "4.5 TEST METHODS".

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4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT.- Each cradle shall be examined visually to determine compliance with the requirements of this specification for which tests are not specified, including materials and workmanship.

4.5.2 CONFORMANCE TO DRAWINGS.- Each cradle shall be inspected to determine conformance to the engineering drawings.

4.5.3 PROOF LOAD TEST.- A 30,000 pound downward vertical load shall be applied to the cradle in a manner simulating the design load. This load shall be applied through the supports, frame and casters. This test will be performed twice, once with the supports in the high position and once with the supports in the low position. Failure or permanent deformation of any part shall be cause for rejection.

4.5.4 ROAD TEST.- The cradle carrying a 10,000 pound test load and loaded on the MHU-7/M Trailer shall be subjected to the type II mobility test described in MIL-M-8090. Failure of any part or appreciable damage shall be cause for rejection.

4.5.5 MOBILITY TEST.- The cradle, carrying a 10,000 pound test load shall be subjected to the type I class I mobility test described in MIL-M-8090 with the following additional procedure:

> With the loaded cradle on level, dry concrete, apply moving force abruptly backwards and continue backwards for a distance of 100 feet at speeds up to 5 mph. The backing procedure shall be repeated five times. Failure, excessive wear or damage to the cradle or any part shall be cause for rejection.

4.5.6 CASTERING TEST.- The trailer carrying a 10,000 pound test load shall be positioned on level, dry concrete. Using a suitable castering device, the front casters shall be rotated 360 degrees and the rear casters rotated a minimum of 180 degrees. At the forward position and at a position of 90 degrees to the forward position, the caster locking provisions shall be applied. Failure of the casters to meet these rotating requirements or failure of the locking provisions shall be cause for rejection.

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4.5.7 FIT TEST.- The cradle shall be assembled in service configuration to the MHU-7/M Trailer, the MB-1 Hoist Tube Set, and the MH-1 and MH-2 Straddle Trucks in turn. The MHU-20/C Clip-In Subassembly shall be mounted on the cradle. Excessive clearance, binding, or misplacement attributable to the location, size, conformation, or finish of attach points of the cradle shall be cause for rejection.

4.5.8 FLIGHT LOAD TEST.- The cradle, without payload, shall be secured in a manner simulating tiedown in a cargo aircraft. While in the air transport configuration, the cradle shall be subjected to the following loads applied in the manner indicated. Direction of application shall be in reference to the aircraft:

a. 3.00g forward, 1.5g upward - applied simultaneously
b. 3.00g forward, 1.5g laterally - applied simultaneously
c. 4.00g forward
d. 1.50g aft
e. 1.50g laterally, left to right, and right to left
f. 3.68g upward

These loads shall be applied once with the front of the cradle toward the front of the aircraft, and once with the rear of the cradle toward the front of the aircraft

g. Operate the lift and brake mechanisms

Failure or permanent deformation of any part shall be cause for rejection.

4.5.9 LOW TEMPERATURE TEST.- The cradle shall be subjected to the low temperature test described in procedure I of MIL-E-4970. Failure to meet the requirements of this test shall be cause for rejection.

4.5.10 ICING TEST.- The cradle shall carry a 10,000 pound test load while being tested in the following manner:

a. Place the cradle in a chamber having a temperature of 49°C (120°F), and relative humidity of 95 percent. These conditions shall be maintained for a period of 48 hours.

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b. Move the cradle into a rain test chamber. A temperature of 20°C (68°F) to 30°C (86°F) shall be maintained in the rain test chamber throughout this portion of the test. A simulated rainfall of 4 ± 1 inch per hour shall be produced by means of a water spray nozzle of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°C (51.8°F) and 20°C (68°F). The rainfall shall be dispersed uniformly over the test area. The cradle shall be subjected to these conditions for a period of 30 minutes.

c. Immediately after subjection to the conditions in step b., the cradle shall be placed in a chamber which has been cooled to a temperature of -18°C (0°F). The trailer shall remain in this environment for a period of 48 hours.

d. Immediately after the exposure in step c., and while at a temperature of -18°C (0°F), the castering provisions shall be operated.

Failure to caster or lock satisfactorily while at -18°C (0°F) shall be cause for rejection.

4.5.11 HIGH TEMPERATURE TEST.- The cradle shall be subjected to the high temperature test described in procedure I of MIL-E-4970. Failure to meet the requirements of this test shall be cause for rejection.

4.5.12 SAND AND DUST TEST.- The cradle shall be subjected to the sand and dust test described in MIL-E-4970. Failure to meet the requirements of this test shall be cause for rejection.

4.5.13 SALT SPRAY TEST.- The cradle shall be subjected to the salt spray test described in MIL-E-4970. Failure to satisfy the requirements of this test shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION.- Preservation for all levels of shipment shall be in accordance with procedure IC, MIL-P-116.

5.2 PACKAGING.- The cradle proper shall constitute a unit package for all levels of shipment. The 4 cable fastener loops and 8 cargo tie down devices shall be packed in a container meeting the requirements of PPP-B-601 and this container secured to the cradle.

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5.3 PACKING

5.3.1 LEVELS A AND B.- For level A and B shipments, the cradle, with accessories secured in accordance with 5.2, shall be physically protected in accordance with JAN-P-100 and packed in an exterior container meeting the requirements of PPP-B-601 or PPP-B-621.

5.3.2 LEVEL C.- The cradle, with accessories secured in accordance with 5.2, shall constitute unit packaging and level C packing.

5.4 MARKING.- Accessory packages and exterior containers shall be marked in accordance with MIL-STD-129. In level C shipment the nameplate of the cradle shall be sufficient marking for the cradle proper. The nomenclature shall be as follows:

Cradle, Bomb Handling MHU-19/E

6. NOTES

6.1 INTENDED USE.- The cradle covered by this specification is intended to be used in the storage, transportation and loading in aircraft of multiple configurations of applicable weapons.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Conditions under which Government-loaned property will be made available to the contractor.

c. Whether preproduction tests will be conducted by the procuring activity or by the contractor under the supervision of the procuring activity.

d. Required levels of preservation, packaging and packing.

6.2.1 PREPRODUCTION TEST SAMPLE.- It is expected that the contract or purchase order will specify that a minimum of one cradle representative of the production cradle will be submitted to the procuring activity for test and evaluation. The invitation for bids and the contract or purchase order should specify the point of inspection for these tests.

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6.3 PRODUCTION DELIVERY.- No production item should be delivered prior to approval of the preproduction samples.

6.4 DEFINITIONS

a. PREPRODUCTION TESTING.- Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order, to prove, prior to his starting production, that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS.- Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine that the units are equivalent in performance and construction to the approved preproduction samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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